



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (MMD) (CBCS) Degree Examinations
May – June 2025

22MMD223 – FRACTURE MECHANICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain how the different failures triggered the fracture study in materials stating any two examples and the lessons from these failure.	10	L*2	01	01
	b) For the DCB as shown in fig 1(b) with thickness B and $a \gg 2h$ and $b \gg 2h$ determine the strain energy release rate and the stress intensity factor at the crack tip using elementary beam theory; Consider as simply supported beam. Comment on the stability of cracking.				

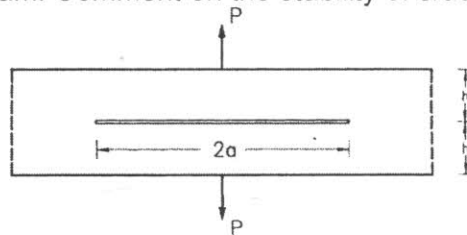


		Fig 1(b)	10	L3	02	02
2.	a)	A sheet of glass 2m long measuring 200mm X 2mm contains a central slit parallel to the 200 mm side. The sheet is restrained at one end and loaded in tension with a mass of 500 kg. What is the maximum allowable length of slit before fracture occurs? Assume plane stress condition and the following material property values: $E = 60\text{GPa}$, Poisson's ratio=0.25, surface energy is 0.5 J/m^2 .	05	L3	01	02
	b)	Differentiate clearly between cleavage fracture and ductile fracture.	05	L2	01	01
	c)	List the various NDT methods used in fracture mechanics. With neat sketch, explain any two NDT method of testing.	10	L3	02	02
3.	a)	Explain the sources of micro and macro cracks, in materials.	04	L2	01	01
	b)	What are the classifications of the different modes of loading at the crack-tip? Explain with neat sketches.	06	L2	01	01
	c)	Indicate graphically the energy availability and derive the energy release rate for crack growth under constant load and constant displacement using Griffith's Criterion.	10	L3	02	02

Unit – II

4.	a)	Derive an expression for the size of the plastic zone and the correction according to Irwin's approach. State also the significance of it.	10	L2	03	01
	b)	Define J-integral. Mention the properties of J-integral and show that it is path independent.	10	L3	04	02

P.T.O.

5. a) A large plate of 5mm thickness made of medium carbon steel $\sigma_{ys}=350\text{MPa}$ with a through thickness center crack of $2a=40\text{mm}$ length subjected to a stress of 150MPa . Find the effective crack length using Irwin's plastic zone correction.
- b) Draw the deformation patterns in plane stress and plane strain loading conditions occurs.
- c) Derive the relationship between CTOD, K_I and G_I .
6. a) Using Dugdale's strip yield model and show that $\rho = (a\pi^2\sigma^2)/(8\sigma_{ys}^2)$.
- b) Discuss the factors affecting the crack propagation.

06	L3	03	02
04	L2	03	01
10	L3	04	01
10	L3	03	02
10	L2	04	01

Unit – III

7. a) A K_{IC} test is performed on a material with $\sigma_{ys}=690\text{MPa}$. The toughness of this material is expected to lie between 40 and $60\text{MPa}\sqrt{\text{m}}$. Design an experiment to measure K_{IC} (in accordance with ASTM E399) in this material using a compact specimen. Specify the following quantities: (a) specimen dimensions, (b) pre-cracking loads, and (c) analyze the required load capacity of the test machine.
- b) Briefly discuss the non-linearity of load-COD diagram.
- c) For fracture toughness testing of metals, Sketch any two ASTM standard test specimens and identify the variables.
8. A three-point bend specimen was tested according to the ASTM E399 procedure. The 0.2 per cent offset yield stress of the material is $\sigma_Y=1200\text{MPa}$ and the modulus of elasticity is $E=210\text{GPa}$. The specimen was tested at a loading rate of 100 kN/min and a type-I load-displacement record was obtained. A chevron starter notch was machined and the specimen was subjected to $30,000$ cycles at $P_{\max}=45\text{kN}$ and $P_{\min}=0$. The final stage of fatigue crack growth was conducted for $50,000$ cycles at $P_{\max}=30\text{kN}$ and $P_{\min}=0$. The specimen dimensions were measured as $S=30\text{cm}$, $W=8\text{cm}$, $B=4\text{cm}$. $a_1=3.996\text{ cm}$; $a_2=4.007\text{ cm}$; $a_3=3.997\text{ cm}$; a (surface) $=3.915\text{ cm}$; a (surface) $=3.952\text{ cm}$. The maximum load and the secant load of the test record were measured as $P_{\max}=86\text{ kN}$ and $P_Q=80\text{ kN}$. Analyze the test outputs and validate the K_{IC} results.

10	L4	05	02
06	L2	05	01
04	L2	05	01
20	L4	05	02

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
